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Teorema fundamental del álgebra

Teorema 1 (Fundamental del álgebra). *El cuerpo $\mathbb{C}$ es algebraicamente cerrado.*

***Demostración*:** Sea $p(z) \in \mathbb{C}[z]$ no constante $(\deg(p(x)) \geqslant 1)$ y suponemos por contradicción que $\forall z \in \mathbb{C} : p(z) \neq 0$. Entonces $f(z) = \frac{1}{p(z)}$ define una función entera. Sabemos que $\lim_{z\to\infty} p(z) = \infty$, es decir,

$$\forall L > 0 : \exists R > 0 : |z| > R \implies |p(z)| > L.$$

Por tanto, $|f(z)| = \frac{1}{|p(z)|} < \frac{1}{L}$, es decir f es acotada.

Entonces, por el teorema de Liouville, f es constante $\implies$ p es constante. $\longrightarrow\!\!\longleftarrow$ ■

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